

Frank — AMC Drill M8 · logic / invariant

25 Jun 2026 | Your real weak spot (M8). Q1–3 = route inspection (twins of 2018 Q24); Q4–6 = cube opposite faces (twins of 2018 Q27). Whole-number answers; show working.

1. [5 marks] A rectangular field is 40 km by 30 km. Fences run along all four sides and along both diagonals, which cross at the centre (each diagonal is 50 km long). A farmer starts at one corner, walks along every fence at least once, and returns to that same corner. What is the shortest total distance, in kilometres, she can walk?

2. [5 marks] A rectangular field is 30 km by 16 km. Fences run along all four sides and along both diagonals, which cross at the centre (each diagonal is 34 km long). A farmer starts at one corner, walks along every fence at least once, and returns to that same corner. What is the shortest total distance, in kilometres, she can walk?

3. [5 marks] A rectangular field is 24 km by 10 km. Fences run along all four sides and along both diagonals, which cross at the centre (each diagonal is 26 km long). A farmer starts at one corner, walks along every fence at least once, and returns to that same corner. What is the shortest total distance, in kilometres, she can walk?

4. [7 marks] Using only the digits 0, 1, 3, each face of a cube shows a different 3-digit number. The three faces meeting at one corner show 100, 130 and 133. The two numbers on each pair of opposite faces add to the same total. What is the largest that this total could be?

5. [7 marks] Using only the digits 1, 2, 4, each face of a cube shows a different 3-digit number. The three faces meeting at one corner show 111, 121 and 124. The two numbers on each pair of opposite faces add to the same total. What is the largest that this total could be?

6. [7 marks] Using only the digits 1, 2, 4, each face of a cube shows a different 3-digit number. The three faces meeting at one corner show 121, 122 and 141. The two numbers on each pair of opposite faces add to the same total. What is the largest that this total could be?

Every item is a code-verified twin of a real 2018 question. Route: count odd junctions first. Cube: the three shown faces are adjacent (meet at a corner), not opposite.